

Use-Case Flow Specification

PES1UG24CS118 | Warehouse Inventory & Pallet Location Tracker

Use Case: UC-02 - Place Pallet

Primary Actor: Warehouse Operator

Preconditions

- The Warehouse Operator is signed in.
- The pallet has a valid barcode or RFID tag.
- The pallet's inventory details already exist in the system.
- A proposed rack and shelf position has been chosen.

Postconditions

- The pallet is assigned to a valid 3D shelf position.
- The pallet's location is saved in the system.
- The rack's load information is brought up to date.
- The stock movement is recorded.

Main Success Scenario

1. The Warehouse Operator scans the pallet's barcode or RFID tag.
2. The system identifies the pallet and pulls up its inventory information.
3. The operator picks a proposed rack and shelf position.
4. The system reads the rack's current load and its maximum permitted load.
5. The system checks the proposed placement against that maximum load capacity.
6. The system confirms the placement is within limits.
7. The system saves the pallet's 3D rack, shelf and slot coordinates.
8. The system logs the movement with the pallet ID, source and destination, timestamp and operator.
9. The system confirms the pallet has been placed successfully.

Alternate Flow - 5a. Rack Capacity Exceeded

- 5a1. The system works out that this placement would take the rack over its maximum rated weight.
- 5a2. The system rejects the proposed placement.
- 5a3. The system tells the operator that the chosen rack cannot take the pallet.
- 5a4. The operator picks a different, valid rack and shelf position.
- 5a5. The use case picks back up from Step 4.